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POWER CONVERSION DEVICE

Abstract

A power conversion device includes a power factor correction circuit includes: a power semiconductor having a heat radiation surface facing a first principal surface of a cooling plate, fixed to the first principal surface so as to exchange heat with the cooling plate, and electrically connected to a drive board; an electrolytic capacitor disposed on an opposite side of the power semiconductor from the cooling plate and a same side of the power semiconductor as the drive board, and electrically connected to the drive board; and a resin mold that is fixed to the first principal surface of the cooling plate, defines a position of the power semiconductor, and is interposed at least between the power semiconductor and the electrolytic capacitor. A second principal surface of the cooling plate is thermally connected to a housing when the power factor correction circuit module is fixed to the housing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-025876, filed Feb. 22, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates to a power conversion device.

BACKGROUND

[0003] In the related art, in a power conversion device such as a charger mounted on an electric vehicle or the like, for example, there is a demand for miniaturization due to limitation of a mounting space.

[0004] For example, Japanese Patent No. 6749428 discloses a technique for miniaturizing an entire device by having a structure in which a power semiconductor, a smoothing capacitor, and a gate drive board are embedded in a power conversion device.

[0005] However, when power conversion is performed on power of a large current or a high voltage, the power loss of the power semiconductor increases, and heat generation from the power semiconductor increases. In addition, with the heat generation from the power semiconductor, the influence of the tailgating heat on the surroundings also increases, and thus, for example, there is a possibility that an adverse effect such as deterioration of the capacitor due to heat may occur. Under such circumstances, in a case of using a liquid-cooled type as a heat radiation method, for example, there was a case where a refrigerant stagnates in a flow path and heat cannot be efficiently dissipated depending on the structure of the flow path, for example, a structure in which a flow path in a vertical direction branches from a flow path in a horizontal direction according to the arrangement of a power semiconductor is employed and the like. Therefore, there was a room for improvement in relation to heat radiation from the power semiconductor.

[0006] One of the problems to be solved by the present disclosure is to miniaturize a power conversion device in consideration of heat radiation properties.

SUMMARY

[0007] A power conversion device according to the present disclosure includes a housing and a power factor correction circuit module. In the housing, a flow path running in one plane is formed. The power factor correction circuit module is detachably fixed to the housing. The power factor correction circuit module includes a cooling plate, a power semiconductor, an electrolytic capacitor, and a resin mold. The cooling plate has thermal conductivity and is formed in a flat plate shape. The power semiconductor has a heat radiation surface facing a first principal surface of the cooling plate, is fixed to the first principal surface so as to be able to exchange heat with the cooling plate, and is electrically connected to a drive board. The electrolytic capacitor is disposed on an opposite side of the power semiconductor from the cooling plate and a same side of the power semiconductor as the drive board, and is electrically connected to the drive board. The resin mold is fixed to the first principal surface of the cooling plate, defines a position of the power semiconductor, and is interposed at least between the power semiconductor and the electrolytic capacitor. A second principal surface that is a back surface of the first principal surface of the cooling plate is thermally connected to the housing in a state where the power factor correction circuit module is fixed to the housing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a diagram illustrating an example of a configuration of a charging system including a power conversion device on which a power factor correction circuit module according to an embodiment is mounted;

[0009] FIG. 2 is a perspective view illustrating an example of a configuration of the power factor correction circuit module of FIG. 1; and

[0010] FIG. 3 is a cross-sectional view illustrating an example of a configuration of the power factor correction circuit module of FIG. 1.

DETAILED DESCRIPTION

[0011] Hereinafter, embodiments of a power factor correction circuit structure, a power factor correction circuit module, a power conversion device, a vehicle, and a charging system according to the present disclosure will be described with reference to the drawings.

[0012] In the description of the present disclosure, components having the same or substantially the same functions as those described above with respect to the previously described drawings are denoted by the same reference numerals, and the description thereof may be appropriately omitted. In addition, even in a case of representing the same or substantially the same portion, the dimensions or ratios may be represented differently from each other depending on the drawings. Furthermore, for example, from a viewpoint of ensuring visibility of the drawings, in the description of each drawing, only main components are denoted by reference numerals, and even components having the same or substantially the same functions as those described above in the previous drawings may not be denoted by reference numerals.

[0013] In the description of the present disclosure, components having the same or substantially the same functions may be distinguished and described by adding alphanumeric characters to the end of reference numerals. Alternatively, in a case where a plurality of components having the same or substantially the same functions is not distinguished, the components may be integrated and described by omitting alphanumeric characters added to the end of the reference numerals.

[0014] FIG. 1 is a diagram illustrating an example of a configuration of a charging system 1 according to an embodiment. As illustrated in FIG. 1, the charging system 1 includes an AC power source 11, a load 13, and a power conversion device 2. FIG. 1 illustrates a case where a three-phase AC power source (external power source) and a battery are connected to the power conversion device 2 as the AC power source 11 and the load 13.

[0015] As an example, the power conversion device 2 according to an embodiment may be mounted on a vehicle as an in-vehicle charger, for example. For example, the power conversion device 2 may be an in-vehicle charger that converts AC power supplied from an external single-phase or three-phase AC power source 11 into DC power and supplies the converted DC power to a load 13 mounted on the vehicle. The load 13 may be, for example, a battery, an inverter, a motor, various electrical components, or the like. In other words, the power conversion device 2 according to the embodiment may be implemented as an in-vehicle charger that converts AC power from the external AC power source 11 into DC power and supplies power to the load 13 such as a battery of a vehicle charged using the DC power.

[0016] As the vehicle, for example, various moving bodies configured to be driven or to be able to drive equipment (electrical equipment) by using electric power from a battery, such as passenger car, freight car, van, motorcycle, electric scooter, and the like can be appropriately used. As the electrical equipment, for example, navigation device, audio device, air conditioner, power window, defogger, electronic control unit (ECU), global positioning system (GPS) module, in-vehicle camera, and the like can be used. In addition, the battery of the vehicle only needs to be able to store electric power for driving a traveling motor (main motor), electric components, and the like mounted on the vehicle, and for example, any battery such as lithium ion battery, nickel hydrogen battery, all-solid battery, and the like can be appropriately used. The power conversion device 2

according to the embodiment is not limited to the vehicle, and may be provided in, for example, an aircraft, a game facility, an uninterruptible power source device, and the like.

[0017] The AC power source **11** is, for example, any external power source such as power source mounted on a rapid charging facility, commercial power source, and the like. The AC power source **11** is not limited to a single-phase AC power source and a three-phase AC power source (multi-phase AC power source), and a two-phase AC power source (multi-phase AC power source) may be used. In the present embodiment, as the AC power source **11** that supplies AC power to the power conversion device **2**, a case where any AC power source **11** of a single-phase or three-phases can be used will be exemplified. That is, in the present embodiment, the power conversion device **2** configured to be operable with both the AC power from the single-phase AC power source **11** and the input of the AC power from the three-phase AC power source **11** will be exemplified.

[0018] As illustrated in FIG. **1**, the power conversion device **2** is electrically connected to the AC power source **11** via any of a plurality of power source lines **L1** to **L3** and **N**. Each of the plurality of power source lines **L1** to **L3** and **N** is electrically connected to an input terminal of the power conversion device **2**. As an example, the power source line **L1** is an electric wire through which a single-phase current from the single-phase AC power source **11** flows. As an example, the power source line **L1** is an electric wire through which a U-phase (first phase) current, for example, from the three-phase AC power source **9b** flows. As an example, the power source line **L2** is an electric wire not electrically connected to the single-phase AC power source **11**. As an example, the power source line **L2** is an electric wire through which a V-phase (second phase) current, for example, from the three-phase AC power source **11** flows. As an example, the power source line **L3** is an electric wire not electrically connected to the single-phase AC power source **11**. As an example, the power source line **L3** is an electric wire through which a W-phase (third phase) current, for example, from the three-phase AC power source **11** flows. As an example, the power source line **N** is a neutral line, and is electrically connected to each of the single-phase or three-phase AC power source **11** and a ground potential.

[0019] As illustrated in FIG. **1**, the power conversion device **2** has a noise filter **4**, a power factor correction (PFC) circuit **5**, and a DC-DC conversion circuit **6**.

[0020] The noise filter **4** suppresses (noise removal) entry of noise from the AC power source **11** to the power conversion device **2** and outflow of noise from the power conversion device **2** to the AC power source **11**. The noise filter **4** is electrically connected to each of the input terminal of the power conversion device **2** and the power factor correction circuit **5** via the plurality of power source lines **L1** to **L3** and **N**. In an example of FIG. **1**, the noise filter **4** has a plurality of coils provided in each of the plurality of power source lines **L1** to **L3** and **N**. Each of the plurality of coils is, for example, a common mode choke coil, a normal mode choke coil, or the like. In addition, a switch that short-circuits between the input terminal of the power conversion device **2** and the power factor correction circuit **5** when turned on is provided in parallel with the coil provided in the power source line **N**. This switch operates, for example, under the control of a control circuit (not illustrated) of the power conversion device **2**.

[0021] Note that, in the noise filter **4**, a plurality of X capacitors may be provided between each coil and the input terminal of the power conversion device **2**. The plurality of X capacitors is electrically connected between each of the power source lines **L1** to **L3** and the power source line **N** (line to line).

[0022] Note that, for example, in a subsequent stage of the noise filter **4**, an inrush preventing resistor may be provided in each of the plurality of power source lines **L1** to **L3**. This inrush preventing resistor is, for example, an inrush current prevention element such as a temperature fuse resistor, a cement resistor, and a thermistor, and prevents an inrush current from flowing through the power factor correction circuit **5**.

[0023] The power factor correction circuit **5** rectifies and smooths the AC voltage from the AC power source **11** to generate a DC voltage. As illustrated in FIG. **1**, the power factor correction

circuit **5** is electrically connected to the noise filter **4** and the DC-DC conversion circuit **6**. The power factor correction circuit **5** has MOSFETs **51a**, **51c**, **51e**, and **51g** on the high side, MOSFETs **51b**, **51d**, **51f**, and **51h** on the low side, and electrolytic capacitors **53a** and **53b**. Here, MOSFETs **51a** to **51h** are examples of a plurality of power semiconductors of the power conversion device **2**. [0024] The MOSFET **51** rectifies the AC voltage from the AC power source **11**. A simple equivalent circuit of each MOSFET **51** is configured using, for example, a switch, a capacitor, and a diode, and operates according to a control signal from a control circuit (not illustrated) of the power conversion device **2**. For example, in the simple equivalent circuit, the switch, the capacitor, and the diode are electrically connected in parallel. For example, in the MOSFET **51** on the high side, an anode and a cathode of the diode are electrically connected to an input side (AC power source **11** side) and an output side (load **13** side) of each MOSFET **51**, respectively. On the other hand, in the MOSFET **51** on the low side, the anode and the cathode of the diode are electrically connected to the output side and the input side of each MOSFET **51**, respectively. In each MOSFET **51**, a switch is turned on/off at a timing corresponding to a control signal from a control circuit (not illustrated) of the power conversion device **2** to switch communication/division. In each MOSFET **51**, charges from an AC power source are accumulated in a capacitor during a period when the switch is turned off (division), and a potential difference occurs between terminals on the input side and the output side. Note that, in the present embodiment, the matter that the internal switch is turned on/off is expressed as that the MOSFET **51** is turned on/off.

[0025] The input side and the output side of each of the MOSFET **51** on the high side are electrically connected to the coil of the noise filter **4** and the input side of each MOSFET **61** on the high side of the DC-DC conversion circuit **6**, respectively. In addition, the input side and the output side of each MOSFET **51** on the high side are electrically connected to the coil of the noise filter **4** and the input side of each of the MOSFET **61** on the low side of the DC-DC conversion circuit **6**, respectively.

[0026] The MOSFETs **51a** and **51b** are examples of a pair of power semiconductors, related to the power source line **L1**, that is, corresponding to a specific phase among a plurality of phases of the AC power source **11**. Specifically, the MOSFETs **51a** and **51b** rectify a single-phase current from the single-phase AC power source **11** flowing through the power source line **L1** or, for example, a U-phase (first phase) current from the three-phase AC power source **11**. MOSFETs **51c** and **51d** are examples of a pair of power semiconductors, related to the power source line **L2**, that is, corresponding to another phase of the specific phase among the plurality of phases of the AC power source **11**. Specifically, the MOSFETs **51c** and **51d** rectify a single-phase current from the single-phase AC power source **11** flowing through the power source line **L2**, or, for example, a V-phase (second phase) current from the three-phase AC power source **11**. The MOSFETs **51e** and **51f** are examples of a pair of power semiconductors, related to the power source line **L3**, that is, corresponding to another phase of the specific phase among the plurality of phases of the AC power source **11**. Specifically, the MOSFETs **51e** and **51f** rectify a single-phase current from the single-phase AC power source **11** flowing through the power source line **L3**, or, for example, a W-phase (third phase) current from the three-phase AC power source **11**. The MOSFETs **51g** and **51h** are examples of a pair of power semiconductors, related to the power source line **N**, that is, corresponding to a neutral line. Specifically, the MOSFETs **51g** and **51h** form an electric circuit for returning the single-phase current flowing through the power source lines **L1** to **L3** from the single-phase AC power source **11** toward the AC power source **11**.

[0027] An electrolytic capacitor **53** smooths the current rectified by the MOSFET **51**. The electrolytic capacitor **53** is an output capacitor of the power factor correction circuit **5**. In addition, the electrolytic capacitor **53** can be expressed as an input capacitor of the DC-DC conversion circuit **6**. FIG. **1** illustrates electrolytic capacitors **53a** and **53b** as the electrolytic capacitor **53**. The electrolytic capacitors **53a** and **53b** are electrically connected between the output side (load **13** side) of the MOSFET **51** on the high side and the output side (load **13** side) of the MOSFET **51** on the

low side. Note that the electrolytic capacitors **53a** and **53b** may be implemented by one electrolytic capacitor, or may be implemented by three or more of a plurality of electrolytic capacitors. For example, FIG. **2** exemplifies a case of being implemented by eight electrolytic capacitors **53**. [0028] The DC-DC conversion circuit **6** converts a DC voltage generated by the power factor correction circuit **5** to an AC voltage again, and then rectifies and smooths the AC voltage to generate a DC voltage of any set voltage. As illustrated in FIG. **1**, the DC-DC conversion circuit **6** is electrically connected to each of the power factor correction circuit **5** and the output terminal of the power conversion device **2**. The DC-DC conversion circuit **6** has MOSFETs **61a** to **61d** on the primary side, a coil **63** on the primary side, a transformer **73**, MOSFETs **81a** to **81d** on the secondary side, a coil **83** on the secondary side, and an output capacitor **85**. Note that each of the coil **63** on the primary side and the coil **83** on the secondary side may be substituted by the leakage inductance of the transformer **73**. Here, the MOSFETs **61a** to **61d** on the primary side are examples of a plurality of power semiconductors of the power conversion device **2**.

[0029] The MOSFETs **61a** to **61d** on the primary side and the coil **63** convert a DC voltage from the electrolytic capacitor **53** as an input capacitor into an AC voltage by a switching operation of the MOSFET **61** under the control of a control circuit (not illustrated) of the power conversion device **2**. That is, the MOSFETs **61a** to **61d** on the primary side and the coil **63** constitute a DC-AC inverter circuit. The simple equivalent circuit of each MOSFET **61** has, for example, the same configuration as that of each MOSFET **51** of the power factor correction circuit **5**. For example, in the MOSFET **61** on the high side, an anode and a cathode of the diode are electrically connected to an output side (load **13** side) and input side (AC power source **11** side) of each MOSFET **61**, respectively. On the other hand, in the MOSFET **61** on the low side, the anode and the cathode of the diode are electrically connected to the input side and the output side of each MOSFET **61**, respectively.

[0030] The input side of each MOSFET **61** on the high side is electrically connected to the output side of each MOSFET **51** on the high side of the power factor correction circuit **5**. In addition, the input side of each MOSFET **61** on the low side is electrically connected to the output side of each MOSFET **51** on the low side of the power factor correction circuit **5**. In addition, the output side of the MOSFETs **61a** and **61b** is electrically connected to one terminal on the primary side of the transformer **73** via the coil **63**. In addition, the output side of the MOSFETs **61c** and **61d** is electrically connected to the other terminal on the primary side of the transformer **73**, that is, the side opposite to the coil **63**.

[0031] The transformer **73** is a potential transformer that transmits energy of the single-phase AC voltage generated by the MOSFETs **61a** to **61d** on the primary side and the coil **63** to the secondary side.

[0032] The MOSFETs **81a** to **81d** on the secondary side and the coil **83** rectify the single-phase AC voltage transmitted by the transformer **73** by a switching operation of the MOSFET **81** under control of the control circuit (not illustrated) of the power conversion device **2**. The simple equivalent circuit of each MOSFET **81** has, for example, the same configuration as that of each MOSFET **51** of the power factor correction circuit **5**. For example, in the MOSFET **81** on the high side, an anode and a cathode of the diode are electrically connected to an input side (AC power source **11** side) and an output side (load **13** side) of each MOSFET **61**, respectively. On the other hand, in the MOSFET **81** on the low side, the anode and the cathode of the diode are electrically connected to the output side and the input side of each MOSFET **81**, respectively.

[0033] The input side of the MOSFETs **81a** and **81b** are electrically connected to one terminal on the secondary side of the transformer **73** via the coil **83**, that is, one terminal having the same polarity as the terminal on the primary side to which the coil **63** is connected in the example of FIG. **1**. In addition, the output side of the MOSFETs **81c** and **81d** is electrically connected to the other terminal on the secondary side of the transformer **73**, that is, the side opposite to the coil **83**. In addition, the output sides of each of MOSFETs **81** on the high side and the low side are

electrically connected to a pair of output terminals of the power conversion device **2**.

[0034] The output capacitor **85** smooths the current rectified by the MOSFET **81**. The output capacitor **85** is electrically connected between an output side (load **13** side) of the MOSFET **81** on the high side and an output side (load **13** side) of the MOSFET **81** on the low side. That is, the output capacitor **85** is electrically connected between the pair of output terminals of the power conversion device **2**.

[0035] Note that a control circuit (not illustrated) of the power conversion device **2** has, for example, at least one processor and at least one memory, and may have a hardware configuration using a normal computer. As this processor, for example, a central processing unit (CPU) can be used. The processor executes, for example, a program to integrally control the operation of the control circuit and implement various functions of the control circuit. Note that the control circuit causes, for example, the processor to load a program stored in a read only memory (ROM) and the like into a random access memory (RAM), and execute the loaded program, and thus the control circuit may implement various functions of the control circuit, or a dedicated hardware circuit (semiconductor integrated circuit and the like) may implement some or all of the various functions.

[0036] Note that the control circuit of the power conversion device **2** may be implemented by a domain control unit (DCU) such as an electronic control unit (ECU) provided inside a vehicle or a cockpit domain controller (CDC) obtained by integrating a plurality of ECUs; or a computer such as an on board unit (OBU). In addition, the control circuit may transmit and receive information with another ECU mounted on the vehicle or an external power source (AC power source **11**) connected to the vehicle via an in-vehicle network including a controller area network (CAN) inside the vehicle, Ethernet (registered trademark), or a universal serial bus (USB) (registered trademark), or may communicate with an information processing device outside the vehicle via a network such as the Internet.

[0037] As illustrated in FIG. **1**, the power factor correction circuit **5** and the MOSFET **61** of the DC-DC conversion circuit **6** constitute a power factor correction circuit module **3** according to the embodiment. That is, the power conversion device **2** according to the embodiment mounts the power factor correction circuit module **3**. Note that the power factor correction circuit module **3** may include other circuit configurations such as the noise filter **4** and the coil **63**, for example.

[0038] Here, the power factor correction circuit structure of the power factor correction circuit module **3** according to the present disclosure will be described in detail with reference to the drawings.

[0039] FIG. **2** is a perspective view illustrating an example of a configuration (power factor correction circuit structure) of the power factor correction circuit module **3** of FIG. **1**. FIG. **2** illustrates the appearance of the power factor correction circuit module **3** of the power conversion device **2** and the vicinity thereof. FIG. **3** is a cross-sectional view illustrating an example of a configuration (power factor correction circuit structure) of the power factor correction circuit module **3** of FIG. **1**. FIG. **3** illustrates a cross-section of the power factor correction circuit module **3** and the vicinity thereof when a Y-Z plane passing through a plurality of fixing members **39c** provided on an X+ side of a relay board **32b** of FIG. **2** is viewed from the X+ side. Note that in FIGS. **2** and **3**, some or all of other configurations of the power conversion device **2** is omitted.

[0040] As illustrated in FIGS. **2** and **3**, the power factor correction circuit module **3** is disposed above (Z+ side) a housing **21** of the power conversion device **2**. As illustrated in FIG. **2**, the power factor correction circuit module **3** is detachably fixed on the housing **21** by a fixing member **39e** such as a screw, for example.

[0041] In addition, as illustrated in FIGS. **2** and **3**, a drive board **31** is disposed above (Z+ side) the power factor correction circuit module **3**. The drive board **31** is a main board of the power conversion device **2**. Each unit of the power conversion device **2** is electrically connected to the drive board **31** directly or via another component. The drive board **31** is detachably fixed to a support unit **25** of the housing **21** by a fixing member (not illustrated) such as a screw.

[0042] Note that the drive board **31** may be a component of the power factor correction circuit module **3**, or may be a component of the power conversion device **2** outside the power factor correction circuit module **3**.

[0043] The housing **21** is a flat plate-shaped member extending along the X-Y plane. Note that the housing **21** is formed by, for example, die casting, but may be formed of iron or may not be made of an alloy. In addition, the housing **21** may be made of another metal material as long as heat from components such as the power factor correction circuit module **3** disposed on the housing **21** can be transported to a coolant flowing through a flow path **23** inside the housing **21**.

[0044] In addition, the housing **21** is provided with a liquid-cooling type cooling mechanism using a coolant such as antifreeze liquid as a working fluid. Specifically, as illustrated in FIG. 3, the flow path **23** of the coolant is formed inside the housing **21**. In the housing **21** according to the embodiment, the flow path **23** extends in a direction (for example, horizontal direction) along the X-Y plane. That is, the flow path **23** that runs in one plane is formed inside the housing **21**.

[0045] The flow path **23** may be branched in at least two directions along the X-Y plane. On the other hand, the flow path **23** according to the embodiment is not branched in a direction along a Z direction. In other words, in the housing **21** according to the embodiment, in a case where there is an intersection where the flow paths **23** in at least two directions intersect, each of the flow paths **23** extends in the direction along the X-Y plane from the intersection, but does not extend in the direction along the Z direction. As described above, the power conversion device **2** according to the present disclosure can constitute a cooling mechanism of the entire system by the flow path **23** running only in one plane (X-Y plane).

[0046] The power factor correction circuit module **3** and the housing **21** are thermally connected. Here, being thermally connected means being configured to be able to exchange heat. Note that heat transport between the power factor correction circuit module **3** and the housing **21** is implemented by, for example, heat conduction, but may be performed in other forms in addition to or instead of heat conduction. In addition, heat transport between the power factor correction circuit module **3** and the housing **21** may be performed via other components.

[0047] In addition, as illustrated in FIG. 2, other components of the power conversion device **2** such as the transformer **73** are also disposed on the housing **21**. The transformer **73** is covered with a casing **75**. The casing **75** is formed of, for example, metal, and shields electromagnetic noise from the transformer **73**. The transformer **73** and the drive board **31** are electrically connected by, for example, a fixing member **77** such as a screw or soldering.

[0048] Generally, in the power conversion device, the transformer **73** is a component (large component) larger than other components. Therefore, the height (length in Z direction) of the support unit **25** of the housing **21** that supports the drive board **31** is determined such that the transformer **73** is accommodated between the housing **21** and the drive board **31**. That is, the height (size in Z direction) of the power conversion device **2** depends on the distance between the housing **21** and the drive board **31** via the transformer **73**. In other words, in a case where the distance (size in Z direction) between the housing **21** and the drive board **31** is defined based on the size of a large component such as the transformer **73**, there is a surplus in height at the disposition position of other components of the large component, that is, at a position different from the large component in an X-Y direction.

[0049] Under such circumstances, the power conversion device **2** according to the present disclosure has the power factor correction circuit module **3** in which some of the other components among the other plurality of components of the transformer **73** are modularized. In addition, the power factor correction circuit module **3** is inserted between the housing **21** and the drive board **31** of which interval is determined in accordance with the transformer **73**. That is, the power factor correction circuit module **3** according to the present disclosure is configured to be smaller in the height direction (Z direction) than other large components of the power conversion device **2**, thereby implementing miniaturization of the power conversion device **2**.

[0050] As illustrated in FIGS. 2 and 3, the power factor correction circuit module 3 further has relay boards 32a and 32b, an insulating heat radiation board 33, and a resin mold 35.

[0051] As illustrated in FIG. 3, the drive board 31 is detachably fixed to the resin mold 35 by a fixing member 39a such as a screw. That is, the resin mold 35 supports the drive board 31.

[0052] Each of the relay boards 32a and 32b is electrically connected to the drive board 31. A wiring pattern is provided on the relay board 32. The relay board 32 electrically connects electrically connected components to the drive board 31 via the wiring pattern. For example, some components of the power factor correction circuit module 3 (power conversion device 2) are electrically connected to the drive board 31 via the relay board 32. As illustrated in FIG. 3, the relay board 32a is detachably fixed to the resin mold 35 by a fixing member 39b such as a screw. In addition, the relay board 32b is detachably fixed to the resin mold 35 by the fixing member 39c such as a screw. That is, the resin mold 35 supports each of the relay boards 32a and 32b.

[0053] Note that the electrical connection between the drive board 31 and the relay board 32 may be implemented by soldering, or may be implemented by detachable connection via a connector.

[0054] As illustrated in FIGS. 2 and 3, the insulating heat radiation board 33 is disposed below (Z-side) the resin mold 35. The resin mold 35 is fixed to a surface (first principal surface) of the insulating heat radiation board 33 on the drive board 31 side by a fixing member (not illustrated) such as a screw and an adhesive. In other words, the resin mold 35 supports the insulating heat radiation board 33.

[0055] The insulating heat radiation board 33 is a member formed in a flat plate shape extending along the X-Y plane. The insulating heat radiation board 33 has thermal conductivity (preferably high thermal conductivity) and electrical insulation properties. Specifically, the insulating heat radiation board 33 only needs to be capable of transporting heat in a thickness direction (Z direction), and is, for example, a metal plate, but may be formed of a non-metal. Here, the insulating heat radiation board 33 according to the embodiment is an example of a cooling plate.

[0056] In addition, an insulating layer having electrical insulation properties is formed on an outer surface of the insulating heat radiation board 33. That is, the insulating layer is provided between each of the MOSFETs 51 and 61 and the insulating heat radiation board 33. Note that the insulating layer on the outer surface of the insulating heat radiation board 33 may be formed by applying coating of a material having electrical insulation properties, or may be formed by providing a sheet-like member having electrical insulation properties on the outer surface. Note that the insulating heat radiation board 33 may be a plate-like member formed of a material having thermal conductivity and electrical insulation properties.

[0057] At least the housing 21 side (Z-side) of the insulating heat radiation board 33 is formed in a shape conforming to the shape of the disposition position of the power factor correction circuit module 3 above (Z+side) the housing 21. Here, a surface of the insulating heat radiation board 33 on the housing 21 side (Z-side) is a cooling surface of the power factor correction circuit module 3. In other words, in the power conversion device 2, the cooling surface of the power factor correction circuit module 3 extends along the plane (X-Y plane) on which the flow path 23 of the housing 21 is provided. That is, the cooling surface of the power factor correction circuit module 3 is thermally connected to the housing 21, so that heat generated inside can be dissipated to the housing 21.

[0058] As an example, the power factor correction circuit module 3 is fixed to the housing 21 by fixing the insulating heat radiation board 33 to the housing 21 with the fixing member 39e. That is, in a state where the power factor correction circuit module 3 is fixed to the housing 21, the surface (second principal surface) of the insulating heat radiation board 33 on the housing 21 side is thermally connected to the housing 21. Here, the surface of the insulating heat radiation board 33 on the housing 21 side is a back surface of the surface (first principal surface) of the insulating heat radiation board 33 on the resin mold 35 side.

[0059] As illustrated in FIG. 3, the MOSFETs 51 and 61 (power semiconductors) of the power

factor correction circuit **5** and the DC-DC conversion circuit **6** are disposed on the surface (first principal surface) of the insulating heat radiation board **33** on the resin mold **35** side. Each of the plurality of MOSFETs **51** and **61** is fixed to the insulating heat radiation board **33** by a fixing member (not illustrated) such as an adhesive or an adhesive sheet. The fixing member may be any fixing member as long as the fixing member is formed of at least a material having thermal conductivity. Note that the fixing member may have electrical insulation properties, and in this case, the insulating heat radiation board **33** may not have electrical insulation properties.

[0060] The cooling surface of each of the plurality of MOSFETs **51** and **61** faces a surface of the insulating heat radiation board **33** on the resin mold **35** side, and is thermally connected to the insulating heat radiation board **33**. That is, the heat radiation path of each of the plurality of MOSFETs **51** and **61** is a path in the Z direction from each of the cooling surfaces to the flow path **23** via the insulating heat radiation board **33** and the housing **21**. As described above, each of the MOSFETs **51** and **61** is fixed to the surface of the insulating heat radiation board **33** on the resin mold **35** side so that heat can be exchanged between the heat radiation surface and the insulating heat radiation board **33**.

[0061] Here, as illustrated in FIG. **3**, each of the plurality of MOSFETs **51** and **61** is flatly placed on the insulating heat radiation board **33**. Here, being flatly placed means that the heat radiation surface of each of the plurality of MOSFETs **51** and **61** is opposed to the surface of the insulating heat radiation board **33** on the resin mold **35** side, and is disposed on the insulating heat radiation board **33**.

[0062] Each of the plurality of MOSFETs **51** and **61** is disposed in a gap **37** formed between the insulating heat radiation board **33** and the resin mold **35**. In other words, each of the plurality of MOSFETs **51** and **61** is disposed below (Z-side) the electrolytic capacitor **53** via the resin mold **35**. For example, each of the plurality of MOSFETs **51** and **61** is disposed immediately below the electrolytic capacitor **53**. That is, an upper side (Z+side) of each of the plurality of MOSFETs **51** and **61** is covered with the resin mold **35**. Each of the plurality of MOSFETs **51** and **61** is defined (positioned) at a position (for example, position in X-Y direction) on the insulating heat radiation board **33** by the resin mold **35**.

[0063] In addition, each of leads **511** and **611** of the plurality of MOSFETs **51** and **61** is electrically connected to the drive board **31**. In the configuration illustrated in FIGS. **2** and **3**, the leads **511** and **611** of each of the plurality of MOSFETs **51** and **61** is electrically connected to one of the relay boards **32a** and **32b**, for example, by soldering.

[0064] As described above, by providing the relay board **32** on the power factor correction circuit module **3**, some of components of the power conversion device **2** such as the MOSFETs **51** and **61** (power semiconductors) can be electrically connected to the drive board **31** via the relay board **32**, and thus a degree of freedom of the disposition can be enhanced. For example, in a case where the MOSFETs **51** and **61** are directly connected to the drive board **31**, there is a limitation in the installation position of the MOSFETs **51** and **61** due to the arrangement of the wire for the connection. On the other hand, in the power factor correction circuit module **3** according to the present disclosure, due to indirect connection to the drive board **31** via the relay board **32**, a degree of freedom of the disposition can be enhanced.

[0065] Note that in some or all of each of the leads **511** and **611** of the plurality of MOSFETs **51** and **61**, the leads **511** and **611** may, for example, extend to the drive board **31** in the Z direction. That is, each of the plurality of MOSFETs **51** and **61** may be electrically connected to the drive board **31** without using the relay board **32**. In this case, some or all of the relay board **32** may not be provided.

[0066] Note that any one of the relay boards **32a** and **32b** may not be provided depending on the number and disposition of the plurality of MOSFETs **51** and **61**. That is, the number of relay boards **32** may be one. On the other hand, the number of relay boards **32** may be three or more.

[0067] As an example, the relay board **32** may be provided on at least one side in the Y direction of

the resin mold **35**. As an example, the relay board **32** may be provided on at least one side in the X direction of the resin mold **35**. That is, the relay board **32** may be provided on at least one side surface portion that extends from a bottom surface portion (part on Z-side) of the resin mold **35** toward the drive board **31** and covers the electrolytic capacitor **53**.

[0068] As an example, the relay board **32** is provided in at least the bottom surface portion (some of Z-side) of the resin mold **35** interposed between the MOSFETs **51** and **61** and the electrolytic capacitor **53**. For example, the relay board **32** may be provided on the side opposite to the insulating heat radiation board **33** of each of the plurality of MOSFETs **51** and **61**. For example, the relay board **32** may be provided between each of the plurality of MOSFETs **51** and **61** and the resin mold **35**. For example, the relay board **32** may be provided below (Z-side) the electrolytic capacitor **53** inside the resin mold **35**. In this case, the relay board **32** may be accommodated inside the resin mold **35** or may protrude from the resin mold **35**.

[0069] As described above, according to the configuration in which the relay board **32** is provided on the side opposite to the insulating heat radiation board **33** of each of the plurality of MOSFETs **51** and **61**, the degree of freedom of the disposition of the plurality of MOSFETs **51** and **61** can be further enhanced. In addition, according to the configuration, radiation heat (air heat) from the plurality of MOSFETs **51** and **61** to other components such as the electrolytic capacitor **53** can be reduced by the relay board **32**.

[0070] In addition, the relay board **32** is not limited to a flat plate-shaped board, and may be a board having an L-shaped cross-section. That is, the relay board **32** may be formed into a shape such as an L-shaped angle (L-shaped angle-iron). That is, the relay board **32** may be a board having at least one L-shaped cross-section supported by at least two of the bottom surface portion and at least one side surface portion of the resin mold **35**. The L-shaped relay board **32** may be provided over two side surfaces (X direction and Y direction) of the resin mold **35**, or may be provided over a side surface (X direction or Y direction) and a bottom surface (Z-side) of the resin mold **35**.

[0071] In addition, the electrolytic capacitor **53** of the power factor correction circuit **5** is disposed on the side (Z+side) of the resin mold **35** opposite to the MOSFETs **51** and **61**, that is, inside the resin mold **35** in the configuration illustrated in FIGS. **2** and **3**. In other words, the electrolytic capacitor **53** is disposed on the drive board **31** side of the MOSFETs **51** and **61**, which is opposite to the insulating heat radiation board **33**.

[0072] A lead **531** of the electrolytic capacitor **53** is electrically connected to the drive board **31**. For example, the electrolytic capacitor **53** is fixed to the drive board **31** by soldering or the like, and is supported by the drive board **31**. As an example, the electrolytic capacitor **53** is disposed in midair from the drive board **31** toward the resin mold **35**. In addition, the electrolytic capacitor **53** may be held by the resin mold **35**, or a position thereof may be defined (positioned) with respect to the drive board **31**. Note that the electrolytic capacitor **53** may be held by the resin mold **35**.

[0073] As described above, the resin mold **35** is interposed at least between the MOSFETs **51** and **61** and the electrolytic capacitor **53**, and radiation heat from the plurality of MOSFETs **51** and **61** and the electrolytic capacitor **53** can be reduced. More specifically, in the power factor correction circuit module **3** according to the present disclosure, the MOSFETs **51** and **61** and the electrolytic capacitor **53** are covered with the resin mold **35**. That is, in the power factor correction circuit module **3** according to the present disclosure, the resin mold **35** plays a role of holding and positioning the MOSFETs **51** and **61** and a role of insulating the electrolytic capacitor **53**.

[0074] Note that the resin mold **35** may further perform positioning of other components such as the electrolytic capacitor **53**, not limited to the MOSFETs **51** and **61**. In other words, the resin mold **35** defines (positions) the position of at least each of the MOSFETs **51** and **61**.

[0075] Note that the gap **37** between the insulating heat radiation board **33** and the resin mold **35** or the gap between the resin mold **35** and the electrolytic capacitor **53** may be filled with a heat radiation buffer such as a gap filler having at least heat radiation properties (thermal conductivity). That is, a heat radiation buffer such as a gap filler may be used as the heat capacity. Accordingly,

the heat radiation performance from the power factor correction circuit module 3 can be further enhanced. The gap filler preferably further has electrical insulation properties.

[0076] A reflective or absorptive heat shielding layer that hinders heat transport to the electrolytic capacitor 53 via the resin mold 35 may be formed on a side of the resin mold 35 facing each of the plurality of MOSFETs 51 and 61. In addition, a reflective heat shielding layer, for example, that hinders heat transfer from the outside may be formed on an outer surface of the electrolytic capacitor 53. These heat shielding layers may be formed, for example, by applying a heat shielding coating to the outer surface of the resin mold 35. Accordingly, heat radiation performance from the power factor correction circuit module 3 can be further enhanced, or deterioration by heat of the electrolytic capacitor 53 can be suppressed.

[0077] Note that the power conversion device 2 according to the present disclosure can be expanded by mounting a plurality of power factor correction circuit modules 3 in parallel. That is, the power conversion device 2 according to the present disclosure has extensibility by modularizing the MOSFET 61 of the power factor correction circuit 5 and the DC-DC conversion circuit 6. For example, the power conversion device 2 can increase the output (kW number) by electrically connecting a plurality of power factor correction circuit modules 3, of a power amount of power conversion, that is, of the number corresponding to the output amount (kW number) of the DC voltage in parallel, that is, by increasing the number of modules.

[0078] As described above, in the power conversion device 2 according to the present disclosure, a power factor correction circuit structure required for power conversion is modularized. Specifically, in the power factor correction circuit module 3 according to the present disclosure, each of the plurality of MOSFETs 51 and 61 can be flatly placed on the insulating heat radiation board 33. Therefore, in the housing 21 of the power conversion device 2, the flow path 23 inside the housing 21 can be formed as a flow path running only in the horizontal direction (X-Y direction), and the stagnation of the working fluid (coolant) can be suppressed. In addition, in the power factor correction circuit module 3 according to the present disclosure, the degree of freedom of the MOSFETs 51 and 61 can be enhanced by using the relay board 32. In addition, by making the power factor correction circuit module 3 according to the present disclosure more compact in a height direction than a large component such as the transformer 73, the entirety of the power conversion device 2 can be miniaturized.

[0079] According to the above-described at least one embodiment, it is possible to miniaturize a power conversion device in consideration of heat radiation properties.

[0080] According to the present disclosure, it is possible to miniaturize a power conversion device in consideration of heat radiation properties.

[0081] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Supplement

[0082] The following techniques are disclosed by the above description of the embodiments.

[0083] (1) [0084] A power conversion device including: [0085] a housing in which a flow path running in one plane is formed; and [0086] a power factor correction circuit module detachably fixed to the housing, wherein [0087] the power factor correction circuit module includes: [0088] a cooling plate having thermal conductivity and formed in a flat plate shape; [0089] a power semiconductor having a heat radiation surface facing a first principal surface of the cooling plate, fixed to the first principal surface so as to be able to exchange heat with the cooling plate, and electrically connected to a drive board; [0090] an electrolytic capacitor disposed on an opposite

side of the power semiconductor from the cooling plate and a same side of the power semiconductor as the drive board, and electrically connected to the drive board; and [0091] a resin mold that is fixed to the first principal surface of the cooling plate, defines a position of the power semiconductor, and is interposed at least between the power semiconductor and the electrolytic capacitor, and [0092] a second principal surface that is a back surface of the first principal surface of the cooling plate is thermally connected to the housing in a state where the power factor correction circuit module is fixed to the housing. [0093] (2) [0094] The power conversion device according to (1), [0095] wherein a lead of the power semiconductor extends to the drive board. [0096] (3) [0097] The power conversion device according to (1), wherein [0098] the power factor correction circuit module further includes a relay board electrically connected to the drive board, [0099] the resin mold supports the relay board, and [0100] the power semiconductor is electrically connected to the relay board and is connected to the drive board via the relay board. [0101] (4) [0102] The power conversion device according to (3), wherein [0103] the resin mold has a bottom surface portion interposed between the power semiconductor and the electrolytic capacitor, and at least one side surface portion extending from the bottom surface portion toward the drive board and covering the electrolytic capacitor, and [0104] the relay board is at least one board supported by at least one of the bottom surface portion and the at least one side surface portion of the resin mold. [0105] (5) [0106] The power conversion device according to (4), [0107] wherein the relay board is a board that is held by at least two of the bottom surface portion and the at least one side surface portion of the resin mold and has at least one L-shaped cross-section. [0108] (6) [0109] The power conversion device according to (3), [0110] wherein the relay board is at least one board that is held by the bottom surface portion of the resin mold interposed between the power semiconductor and the electrolytic capacitor. [0111] (7) [0112] The power conversion device according to any one of (1) to (6), wherein [0113] the power semiconductor is disposed in a gap between the first principal surface of the cooling plate and the resin mold, and [0114] the gap is filled with a heat radiation buffer. [0115] (8) [0116] The power conversion device according to any one of (1) to (7), [0117] wherein the power factor correction circuit module further includes the drive board. [0118] (9) [0119] The power conversion device according to any one of (1) to (8), wherein [0120] the power factor correction circuit module includes a number of power factor correction circuit modules, the number corresponding to a power amount of power to be converted. [0121] (10) [0122] The power conversion device according to any one of (1) to (9), further including [0123] a transformer that is fixed to the housing and electrically connected to the drive board, [0124] wherein the power factor correction circuit module is smaller than the transformer in a direction from the housing toward the drive board. [0125] (11) [0126] A vehicle including: [0127] a power conversion device according to any one of (1) to (10), which converts AC power from an external AC power source to DC power; and [0128] a battery charged using the DC power converted by the power conversion device.

Claims

1. A power conversion device comprising: a housing in which a flow path running in one plane is formed; and a power factor correction circuit module detachably fixed to the housing, wherein the power factor correction circuit module includes: a cooling plate having thermal conductivity and formed in a flat plate shape; a power semiconductor having a heat radiation surface facing a first principal surface of the cooling plate, fixed to the first principal surface so as to be able to exchange heat with the cooling plate, and electrically connected to a drive board; an electrolytic capacitor disposed on an opposite side of the power semiconductor from the cooling plate and a same side of the power semiconductor as the drive board, and electrically connected to the drive board; and a resin mold that is fixed to the first principal surface of the cooling plate, defines a position of the power semiconductor, and is interposed at least between the power semiconductor and the electrolytic capacitor, and a second principal surface that is a back surface of the first

- principal surface of the cooling plate is thermally connected to the housing in a state where the power factor correction circuit module is fixed to the housing.
2. The power conversion device according to claim 1, wherein a lead of the power semiconductor extends to the drive board.
 3. The power conversion device according to claim 1, wherein the power factor correction circuit module further includes a relay board electrically connected to the drive board, the resin mold supports the relay board, and the power semiconductor is electrically connected to the relay board and is connected to the drive board via the relay board.
 4. The power conversion device according to claim 3, wherein the resin mold has a bottom surface portion interposed between the power semiconductor and the electrolytic capacitor, and at least one side surface portion extending from the bottom surface portion toward the drive board and covering the electrolytic capacitor, and the relay board is at least one board supported by at least one of the bottom surface portion and the at least one side surface portion of the resin mold.
 5. The power conversion device according to claim 4, wherein the relay board is a board that is held by at least two of the bottom surface portion and the at least one side surface portion of the resin mold and has at least one L-shaped cross-section.
 6. The power conversion device according to claim 3, wherein the relay board is at least one board that is held by the bottom surface portion of the resin mold interposed between the power semiconductor and the electrolytic capacitor.
 7. The power conversion device according to claim 1, wherein the power semiconductor is disposed in a gap between the first principal surface of the cooling plate and the resin mold, and the gap is filled with a heat radiation buffer.
 8. The power conversion device according to claim 1, wherein the power factor correction circuit module further includes the drive board.
 9. The power conversion device according to claim 1, wherein the power factor correction circuit module includes a number of power factor correction circuit modules, the number corresponding to a power amount of power to be converted.
 10. The power conversion device according to claim 1, further comprising a transformer that is fixed to the housing and electrically connected to the drive board, wherein the power factor correction circuit module is smaller than the transformer in a direction from the housing toward the drive board.
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